

PROJECT PRESENTATION

AI Pseudonymizer for Medical Data

A local middleware that lets external AI analyse clinical documents - without ever exposing patient identity.

n8n pipeline · Hybrid Regex + local LLM detection · External LLM analysis
Data Security & Privacy module · June 2026

The problem and our solution on one page

THE DILEMMA

AI brings huge clinical value

Summarising histories, decision support, ICD coding...

...but it needs powerful external models

And sending raw clinical data out breaches the GDPR.

Today the choice is binary

Drop AI to stay private, or send data unprotected and break the law.

OUR SOLUTION

A local pseudonymization middleware

Detects PII, swaps it for plausible fakes, lets the external LLM work on the safe text, and restores identity at the end.

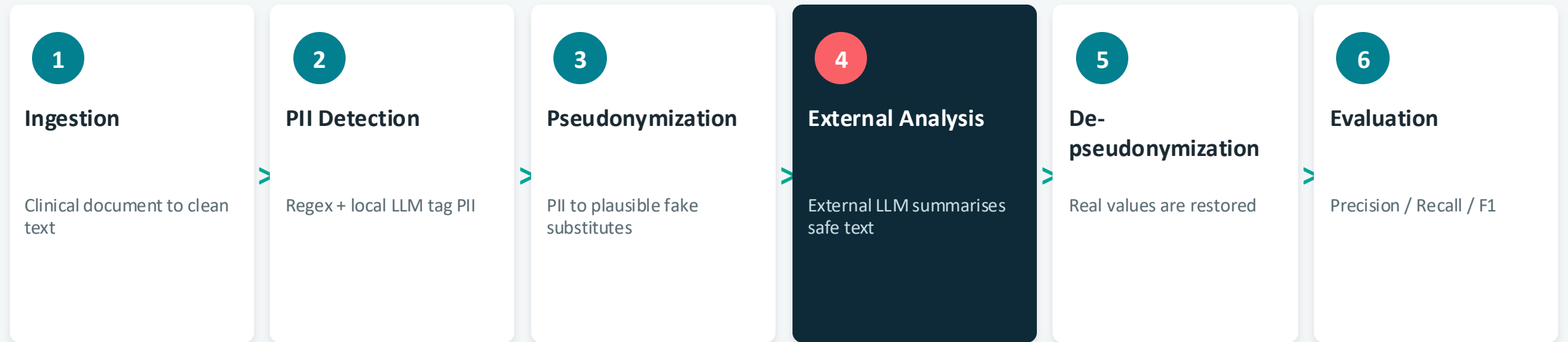
PII never leaves unprotected

Detection, pseudonymization and restoration stay in the local environment.

Keep the value, remove the risk

A principled middle ground, GDPR-aligned by design.

The end-to-end flow



stages 1-3, 5 and 6 run locally. Only the **already-pseudonymized** text crosses to the external provider in stage 4.

Architecture: which model does what

LOCAL ENVIRONMENT · real data

PII Detector - AI Agent (n8n)

Model: Ollama · mistral:latest (Mistral 7B)

Local base URL (host.docker.internal:11434)

Regex pre-pass (German clinical)

Dates, ages, IDs, addresses, titled names and physician signatures.

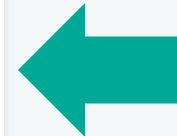
Pseudonymization + restoration

In-memory dictionary; never persisted to disk.

pseudonymized
text



analysis
returned



EXTERNAL PROVIDER · fake data only

Clinical analyst - AI Agent1 (n8n)

Model: Mistral Cloud (La Plateforme)

OpenAI-compatible endpoint

Writes the Epikrise / Verlaufsbericht

Works on fake identities; never sees real patient data.

Why split the models: the detector sees real PHI, so it must be local; the summariser only sees pseudonyms, so it can use a powerful external LLM. Data minimisation by design.

The n8n workflow

1 Start to Load Input Loads the GraSCCo corpus (UIMA CAS JSON), extracts text + ground-truth PII

2 A • Detect (Regex) Deterministic German-clinical detection

3 AI Agent • Ollama Local LLM detects PII in free text

4 Merge Detections Combines regex + LLM, filters hallucinations and duplicates

5 B • Pseudonymize Swaps PII for fake values; builds the dictionary

6 AI Agent1 • Mistral Cloud External LLM writes the clinical summary

7 C • De-pseudonymize Reverts pseudonyms using the dictionary

8 D • Evaluate Computes precision / recall / F1 vs ground-truth

Evaluation loop

Loop Over Items (batch 1)
processes documents one by one.

Code • Aggregate
micro-averages TP/FP/FN across the corpus into one citable result.

Icons:
green = local · red = external

What we defined as PII

Names

Patients, physicians, staff and relatives - incl. lone initials ("M.") and ALL-CAPS ("H. BLASENSTEIN").

Dates

Birth, admission, discharge; German formats (1.2.2000, 3.5.) and standalone years.

Age

Numeric age of a person (e.g. "28").

Addresses

Streets + number, postal code, city, clinics and universities.

Contact

Phone, fax, email.

ID numbers

Case, insurance, patient or document numbers.

Profession

Occupation of patient or relatives (e.g. "Floristin").

Titles

Dr., Dr. med., Prof., PD - included even when standalone.

PII Detector prompt (local LLM)

AI Agent · Ollama mistral:latest · system message

```
You are a strict clinical data-protection assistant (EU GDPR) for German
medical texts. Extract EVERY personal identifier. Favor recall.
```

EXTRACT:

```
NAME      patients, physicians, relatives; bare surnames; ALL-CAPS; initials ("M.")
TITLE     Dr., Dr. med., Prof., PD, Priv.-Doz. -- even when alone
DATE      dates and standalone years (2006, 3.5., 8.5.2028)
AGE       numeric age of a person ("28")
ADDRESS   streets+number, postal code, city, clinic/university, ward
PROFESSION occupation of patient/relative ("Floristin")
CONTACT   phone, fax, email      ID_NUMBER case/insurance/patient numbers
```

RULES:

1. Copy each value EXACTLY (verbatim substring). Never translate or invent.
2. Split comma-separated name lists into separate entities.
3. For "Title + Name" output BOTH the full form AND the bare name.

```
NEVER EXTRACT: diagnoses, anatomical terms / vertebral levels (L4/5),
department or treatment descriptions, measurements, lab values, drug
names/doses, placeholders ("none", <angle brackets>).
```

```
Return ONLY valid JSON:
```

```
{"entities":[{"type":"NAME|TITLE|DATE|AGE|ADDRESS|PROFESSION|CONTACT|ID_NUMBER",
"value":"<verbatim>"}]}
```

Clinical Summariser prompt (external LLM)

AI Agent1 · Mistral Cloud · system message · input = pseudonymized text

You are an expert clinical documentation assistant. Your task is to write a concise, professional medical summary (Epikrise / Verlaufsbericht) of the provided clinical note.

CRITICAL PROCESSING LAWS:

1. VALUE PRESERVATION -- copy EVERY name, date and age VERBATIM from the source. Never paraphrase or drop specific values.
2. NO HALLUCINATION -- do not add any medical fact, diagnosis, symptom or history that is not explicitly stated in the source text.
3. MEDICAL RESTRUCTURING -- you may improve sentence flow, but only using facts present in the source.
4. LANGUAGE -- German, professional Fachsprache.
5. FORMAT -- concise structured bullet points.

The value-preservation rule is what lets de-pseudonymization (stage 5) reliably recover every identity.

RESULTS

Measured performance on the corpus

Micro-average over 5 GraSCCo clinical notes · 100 ground-truth PII entities

PII DETECTION (Regex + local LLM)

0.921

Precision

0.820

Recall (coverage)

0.868

F1-score

RESTORATION (de-pseudonymization, over what appears in the summary)

0.942

Recall

0.845

Precision

0.891

F1-score

82 of 100 identifiers detected · 7 false positives · near-perfect restoration of the entities the summary uses.

Regex only vs Regex + local LLM

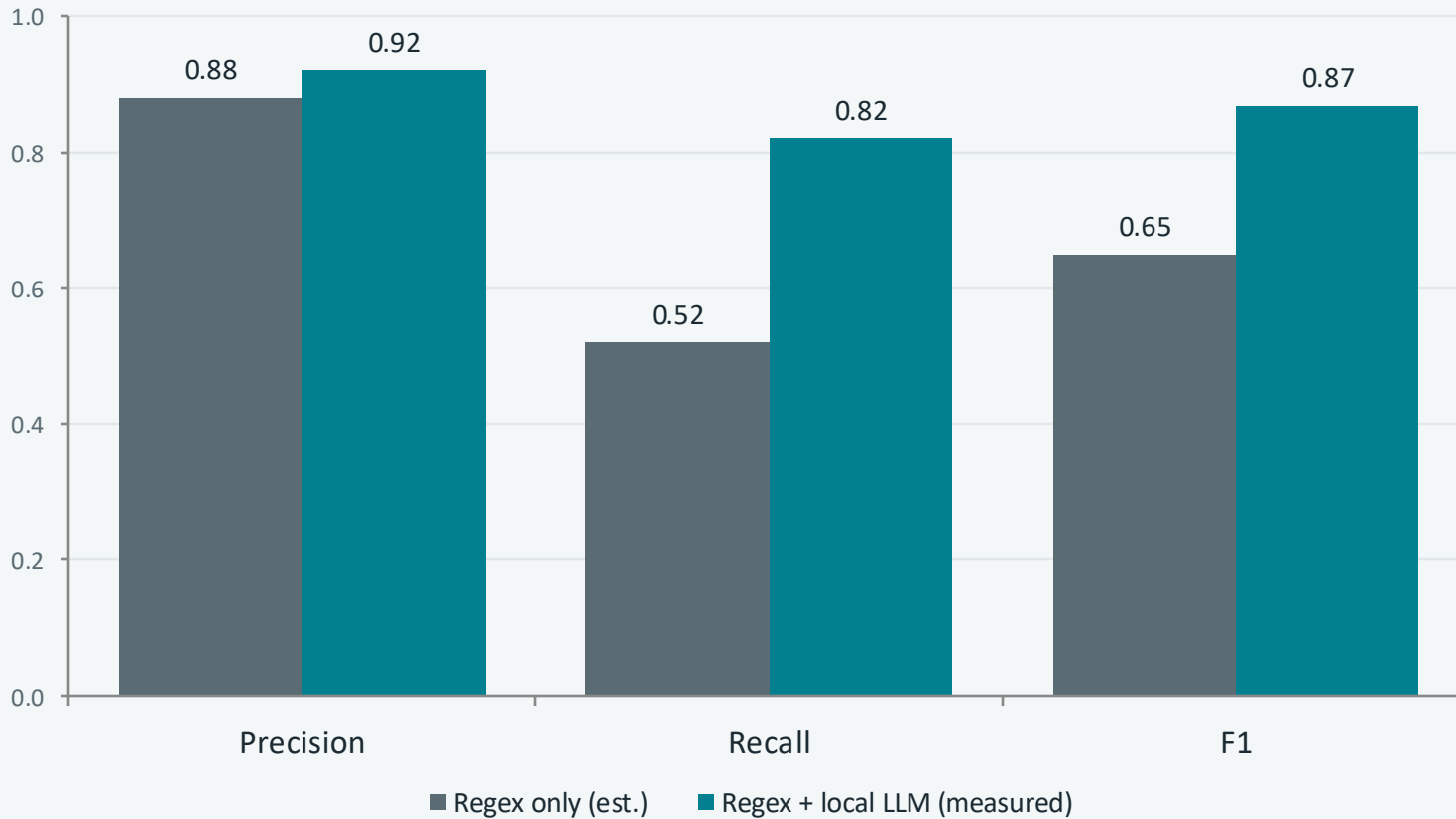
	Regex only	Regex + local LLM (chosen)
Approach	Pattern rules for IDs, dates, formats	Regex pre-pass + LLM reads context
Speed	Fastest - near-instant	Moderate - model inference per doc
Free-text names	Misses unstructured ones	Catches them by context
Consistency	Fully deterministic	Stable, regex-anchored; LLM varies a bit
Precision / Recall	High precision, low recall	Best balance: P 0.92 · R 0.82
Best for	Structured fields, baseline	Production privacy guarantee

Our choice: the hybrid. Regex gives speed and precision on structured fields; the local LLM recovers the free-text names regex misses - coverage is what prevents leaks.

RESULTS

Regex vs Regex + local LLM - the numbers

Detection scores by metric · higher is better



What the chart shows

Precision stays high

0.88 → 0.92. Both approaches rarely flag non-PII.

Recall is the big jump

0.52 → 0.82 (+0.30). The local LLM catches the free-text names regex misses.

F1 follows

0.65 → 0.87. Coverage is what closes the leak gap.

CONCLUSION

Keep the AI value, remove the privacy risk

1

Outcome

An end-to-end n8n pipeline that analyses documents via external AI with PII protected throughout.

2

Practical value

Lets healthcare facilities adopt high-value clinical AI without breaching the GDPR.

3

Measured & extensible

Objective precision/recall scoring, a provider-agnostic gateway, and room for more templates.

Thank you

Questions on the architecture, the privacy model, or the evaluation are welcome.